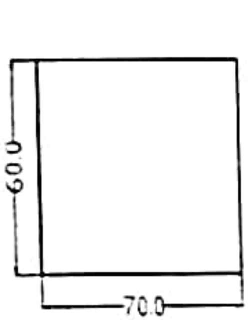
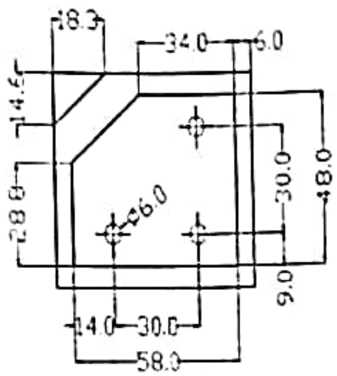
(b) Raw Material   Finished Part





8. Write short notes on: (any four)

a) Robotics and Mechatronics
b) Flexible Manufacturing Systems
c) Manual and Computer Assisted (APT) CNC Part Programming
d) Computer Integrated Manufacturing Systems
e) Plastic Deformation and Yielding Criteria
f) Hot and Cold Working
g) Friction and Lubrication in Metal Forming

4x4=16

"Do not write anything on question paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules"

Roll No.: 20BMEC1162

BE IV (M)
3

Manu. Tech.

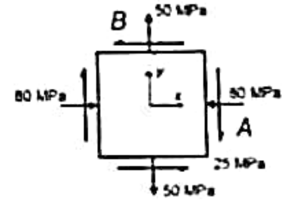
FINAL B.E. EXAMINATION, 2023
(MECHANICAL ENGINEERING)
(VII SEMESTER CBCS)

ME 404 A :MANUFACTURING TECHNOLOGY

Time – Three Hours
Maximum Marks – 80

Note: (i) Attempt any Five questions.
(ii) Answer should be to the point.
(iii) All questions carry equal marks.

1. (a) Find magnitudes and directions of principal stresses, maximum shearing stress and corresponding normal stress considering plane stress system shown below. Also construct Mohr's circle for the same. 10



(b) For problem given in question 1(a), find yield stress of the material according to:
(i) Tresca criteria
(ii) Von-Mises Criteria 6

2. (a) Explain rolling process and classify it. Derive expression for roll load, torque and mill power for cold rolling. 10

Cont.

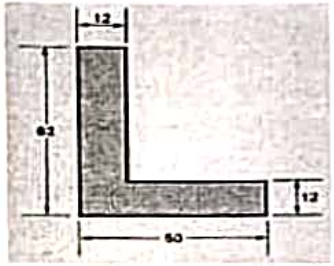
(b) A 2.0 inch thick slab is 10.0 inch wide and 12.0 feet long. Thickness is to be reduced in three steps in a hot rolling operation. Each step will reduce the slab to 75% of its previous thickness. It is expected that for this metal and reduction, the slab will widen by 3% in each step. If the entry speed of the slab in the first step is 40 feet/min, and roll speed is the same for the three steps, determine: (i) length and (ii) exit velocity of the slab after the final reduction. 6

3. (a) Differentiate between the various forging processes and derive expression for maximum force required for forging. 10

(b) A connecting rod is designed to be hot forged in an impression die. The projected area of the part is 6,500 mm^2. The design of the die will cause flash to form during forging, so that the area, including flash, will be 9,000 mm^2. The part geometry is considered to be complex. As heated the work material yields at 75 MPa, and has no tendency to strain harden. Determine the maximum force required to perform the operation assuming K_f=8.0 6

4. (a) Distinguish between direct and indirect extrusion. Why is friction a factor in determining the ram force in direct extrusion but not a factor in indirect extrusion? 10

(b) An L-shaped structural section is direct extruded from an aluminum billet in which L_0 = 500 mm and D_0 = 100 mm. Dimensions of the cross section are given in below figure. Die angle = 90°. Determine (a) extrusion ratio, (b) shape factor, and (c) length of the extruded section if the butt remaining in the container at the end of the ram stroke is 25 mm. 6



5. (a) For the following applications, identify one or more nontraditional machining processes that might be used, and present arguments to support your selection.
(i) A through-hole in the shape of the letter L in a 12.5mm thick plate of glass. The size of the "L" is 25 x 15mm and the width of the hole is 3 mm.
(ii) A blind-hole in the shape of the letter G in a 50 mm cube of steel. The overall size of the "G" is 25 x 19mm, the depth of the hole is 3.8 mm, and its width is 3 mm. 10

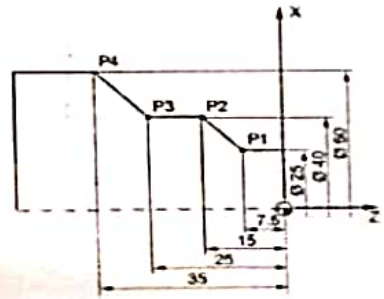
(b) Differentiate between various modern machining methods on the basis of energy utilized by them. 6

6. (a) Describe various methods of producing gears. 8

(b) Explain Rapid Prototyping processes and their applications. 8

7. Write CNC part program for following: 8+8=16

(a)



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Roll No. 21UMEC1122

BE IV (ME)

3 **MT**

FINAL B.E. EXAMINATION - 2024

MECHANICAL ENGINEERING

(VII SEMESTER CBCS)

ME - 404 A : MANUFACTURING TECHNOLOGY

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) Considering plane stress system, find magnitudes and directions of principal stresses, maximum shearing stress and corresponding normal stress. Also construct Mohr's circle for $\sigma_x = 750\ N/mm^2$; $\sigma_y = 150\ N/mm^2$; $\sigma_z = 0$ and $\tau_{xy} = 150\ N/mm^2$ 10

6

(Contd.)

(b) For problem given in question 1(a), find yield stress of the material according to:

(a) Tresca criteria

(b) Von-Mises Criteria 6

2. (a) What is rolling in the context of the bulk deformation processes? Identify some of the ways in which force in flat rolling can be reduced. What is draft in a rolling operation? 10

6/04

12

(b) A series of cold rolling operations are to be used to reduce the thickness of a plate from 50 mm down to 25 mm in a reversing two-high mill. Roll diameter = 700 mm and coefficient of friction between rolls and work = 0.15. The specification is that the draft is to be equal on each pass. Determine (a) minimum number of passes required, and (b) draft for each pass? 6

3. (a) What is forging? One way to classify forging operations is by the degree to which the work is constrained in the die. By this classification, name

(Contd.)

the three basic types. What are the basic types of forging equipment? 10

(b) A hot upset forging operation is performed in an open die. The initial size of the workpart is: D_o=25mm, and h_o = 50mm. The part is upset to a diameter = 50 mm. The work metal at this elevated temperature yields at 85 MPa (n = 0). Coefficient of friction at the die-work interface = 0.40. Determine (a) final height of the part, and (b) maximum force in the operation. 6

4. (a) Highlight the importance of extrusion process and derive expression for frictional power loss and maximum force required in extrusion process. 10

(b) A 3 inch long cylindrical billet whose diameter =1.5 inch is reduced by indirect extrusion to a diameter = 0.375 inch. Die angle = 90°. In the Johnson equation, a = 0.8 and b = 1.5. In the flow curve for the work metal, K = 75,000 lb/inch^2 and n = 0.25. Determine (a) extrusion ratio, (b) true

(Contd.)

strain (homogeneous deformation), (c) extrusion strain, (d) ram pressure, (e) ram force, and (f) power if the ram speed = 20 in/min. 6

5. (a) Compare Electrochemical Machining and Electric Discharge Machining in terms of operating principle, key elements and process parameters. 8

6

(b) Compare Electron Beam Machining and Laser Beam Machining in terms of operating principle, key elements and process parameters. 8

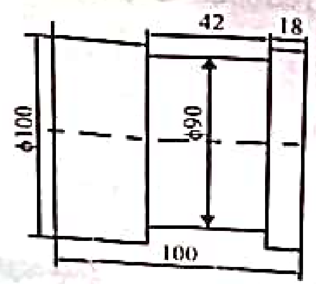
6. (a) Give details of three basic sheet metal processes: Cutting, Bending and Drawing with suitable examples. 8

(b) Explain Flexible Manufacturing Systems and their applications. 8

7. Write CNC part program for following :

10

(a)



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